

CHEST PAIN

Give urgent attention to the patient with chest pain and any of:

- Respiratory rate ≥ 30 or difficulty breathing
- BP $\geq 180/110$ or $< 90/60$
- Pulse irregular, > 100 or < 50
- Severe pain
- New pain or discomfort in centre or left side of chest
- Pain radiates to neck, jaw, shoulder/s or arm/s
- Nausea or vomiting
- Pallor or sweating
- Known with ischaemic heart disease
- At risk of heart attack (diabetes, smoker, hypertension, high cholesterol, known CVD risk $> 20\%$, family history)

Do an ECG.

ECG abnormal
(ST elevation, ST depression or left bundle branch block)

ECG normal/other abnormalities or unavailable or uncertain

Is chest pain worse on lying down, palpation or breathing deeply?

No

Ischaemic heart disease likely → 137.

Yes

Manage and refer urgently:

- If oxygen saturation $< 94\%$, oxygen saturation not available, respiratory rate ≥ 30 or difficulty breathing, give face mask oxygen.
- If sudden breathlessness, more resonant/decreased breath sounds/pain on one side, deviated trachea: **tension pneumothorax** likely:
- Doctor to insert large bore cannula above 3rd rib in mid-clavicular line and arrange urgent chest tube.
- If BP $< 90/60$, give **sodium chloride 0.9%** 500mL IV over 30 minutes, repeat until systolic BP > 90 . Continue 1L 6 hourly. Stop if breathing worsens.
- If BP $> 180/130$, give single dose **amlodipine** 10mg orally.
- If chest pain worse on breathing deeply, coughing sputum and temperature $\geq 38^\circ\text{C}$, give **ceftriaxone** 1g IV¹/IM to cover for severe pneumonia.

Approach to the patient with chest pain not needing urgent attention:

- If recurrent episodes of central chest pain, brought on by exertion and relieved by rest, **ischaemic heart disease** likely → 137.
- If cough, fever or pain on breathing deeply → 38.
- If ongoing chest pain after an acute COVID-19 infection → 42.
- Ask about site of pain and associated symptoms:

Retrosternal or epigastric pain with eating, hunger or lying down/bending forward

Dyspepsia (heartburn) likely

- Advise to stop NSAIDs (ibuprofen/aspirin), quit smoking → 141, limit alcohol, caffeine, spicy food, fizzy drinks, late night meals.
- If waist circumference $> 80\text{cm}$ (woman) or 94cm (man), assess CVD risk → 127.
- Give **lansoprazole**² 30mg daily for up to 14 days.
- Refer same week if any of: no better after 7 days treatment, symptoms return, painful/difficulty swallowing, persistent vomiting, abdominal mass, blood in vomit or stool (occult blood positive), weight loss, Hb < 12 (woman) or < 13 (man), new pain and > 50 years, or family history of stomach/oesophageal cancer.

Tender at costochondral junction, no fever or cough

Musculoskeletal problem likely

- Give **ibuprofen** 400mg 8 hourly with food for up to 5 days (avoid if peptic ulcer, asthma, hypertension, heart failure or kidney disease).
- If pain persists > 4 weeks, refer.

Burning pain on one side of body with or without rash

Herpes zoster (shingles) likely → 68.

If diagnosis uncertain, refer same week.

¹Do not mix Ringer's lactate and IV ceftriaxone. Flush IV line with **sodium chloride 0.9%** before and after IV ceftriaxone. ²Avoid if on atazanavir/ritonavir. Discuss with specialist.